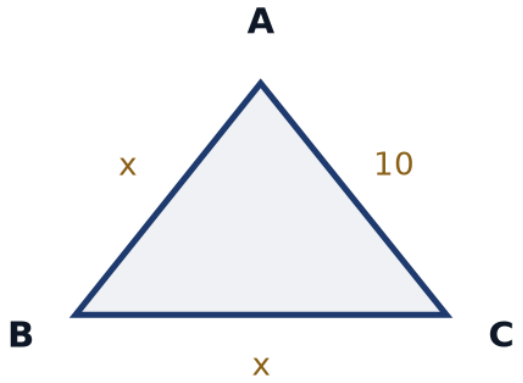
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Two angles, $\angle A$ and $\angle B$, are complementary. If $\angle A = (3x + 15)^\circ$ and $\angle B = (2x + 30)^\circ$, what is the measure of $\angle A$?

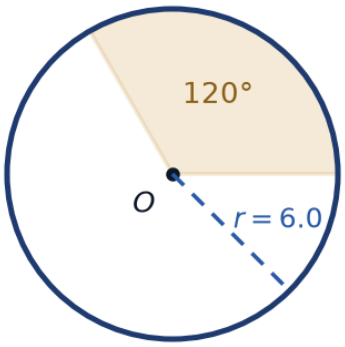
2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In the isosceles triangle ABC shown below, AB = AC. If the measure of angle BAC is 52°, what is the measure of angle ABC?



3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In the given circle, the central angle $\angle AOB$ measures 120°. What is the measure of the inscribed angle $\angle ACB$ that subtends the same arc AB?



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4 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider the line L with the equation $3x - 4y + 15 = 0$. A point P has coordinates $(5, -2)$. Calculate the shortest perpendicular distance from point P to line L.

5 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In ΔABC, points M and N are the midpoints of sides AB and AC respectively. If the area of ΔAMN is $18\ \text{cm}^2$, what is the area of trapezoid $BMNC$?

6 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In the figure below, lines AB and CD are parallel with $AB = 10$ and $CD = 15$. Segments AD and BC intersect at point E. If the perimeter of ΔABE is 26, what is the perimeter of ΔCDE?

7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Chords AB and CD intersect inside a circle at point P. The segments on chord AB measure $PA = x$ and $PB = x + 7$. The segments on chord CD measure $PC = 3$ and $PD = 6$. What is the total length of chord AB?

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In ΔABC, $\angle A = 50^\circ$. Transversal DE is drawn such that $\angle ADE = \angle C = 65^\circ$. What is the measure of $\angle B$?

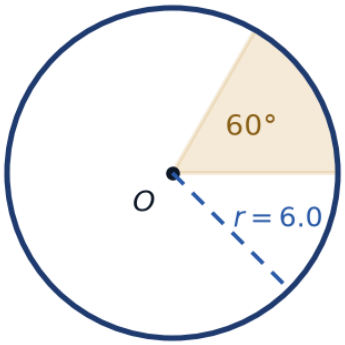
9 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In an hourglass figure with $AB \parallel CD$, $\angle BAE = 38^\circ$ and $\angle AEB = 84^\circ$. What is the measure of $\angle EDC$?

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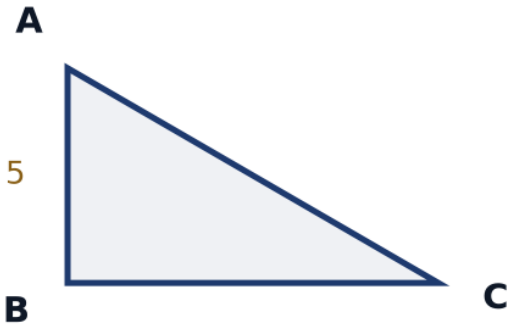
10 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A circle has a radius of 6. A sector of the circle is formed by a central angle of 60°. What is the area of the sector?



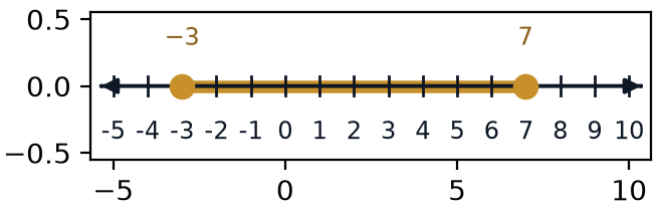
11 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In right triangle ABC, $\angle B = 90^\circ$, $\angle A = 60^\circ$, and the length of side AB is 5. What is the length of side BC?



12 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

On a number line, point A is at -3, and point B is at 7. Point M is the midpoint of segment $\overline{AB}$. What is the coordinate of point M?



13 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In the given right triangle ABC, where angle C is the right angle, the length of side AC is 5 units and the length of side BC is 5 units. What is the exact length of the hypotenuse AB?

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14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In ΔABC, line segment DE is parallel to BC, with point D lying on $\overline{AB}$ and point E lying on $\overline{AC}$. If $AD = 3$, $DB = 6$, and $AE = 5$, what is the length of segment $\overline{EC}$?

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 4.01 Similarity & Proportional Reasoning

In the coordinate plane, the directed line segment from $A(-5, -3)$ to $B(5, 7)$ is partitioned by point P in the ratio $3 : 2$. What are the coordinates of point P?

16 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 5.01 Right Triangles & Trigonometry

A student whose eyes are 1.5 m above the ground places a mirror flat on the ground 2 m in front of her feet. Looking into the mirror, she sees the reflection of the top of a school flagpole located 18 m beyond the mirror. If the angle of incidence equals the angle of reflection, how tall is the flagpole?

17 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 6.01 Polygons & Quadrilaterals

In ΔABC, line segment DE is drawn parallel to base BC such that it divides the triangle into two regions of equal area: $\text{Area}(\Delta ADE) = \text{Area}(BDEC)$. What is the ratio $\frac{AD}{AB}$?

18 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 7.01 Circles

In ΔABC, a transversal segment DE connects side AB at D and side AC at E such that $\angle ADE = \angle C$. If $AD = 3$, $AC = 9$, and the area of ΔADE is 8, what is the area of quadrilateral $BDEC$?

19 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 8.01 Area, Surface Area & Volume

In ΔABC shown below, point D is on AB and E on AC such that $\angle ADE \cong \angle C$. If $AD = 4$, $AE = 5$, and $AC = 12$, what is the length of side AB?

20 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 9.01 Transformations & Symmetry

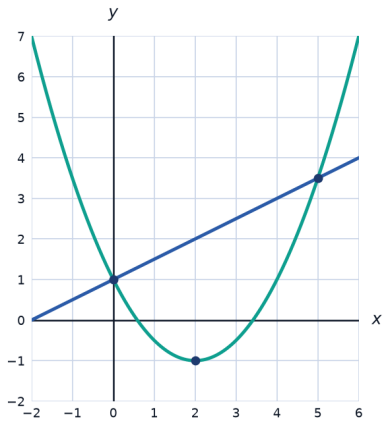
ΔABC and ΔDEF are shown below with $\angle A \cong \angle D = 55^\circ$. If $AB = 6$, $AC = 10$, $DE = 9$, and $DF = 15$, which similarity criterion applies and what is the scale factor from ΔABC to ΔDEF?

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 2x + 3$ in points, lines, and planes, evaluate $f(2)$.



a. 7

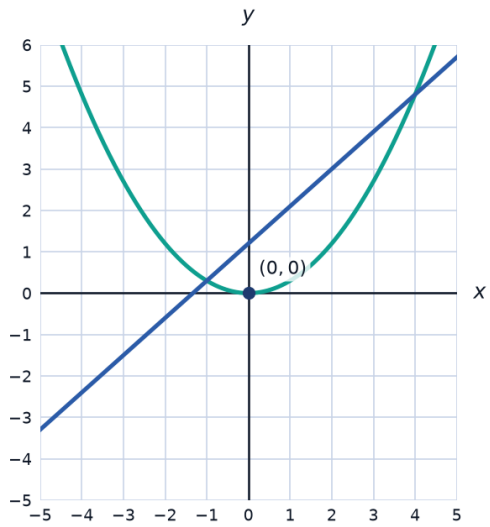
b. 9

c. 4

d. 11

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the positive solutions to the quadratic equation $x^2 - 6x + 8 = 0$.



a. $x = 2$ and $x = 5$

b. $x = 2$ and $x = 4$

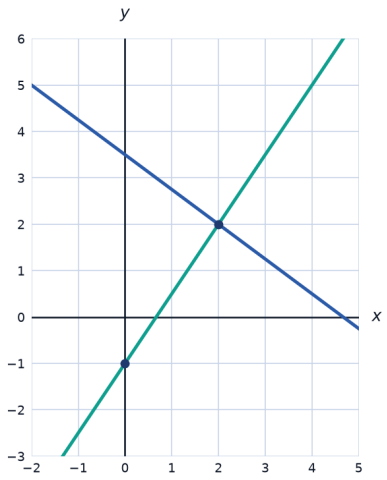
c. $x = -2$ and $x = -4$

d. $x = 6$ and $x = 8$

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23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 3x + 4$ in line segments and distance, evaluate $f(2)$.



a. 10

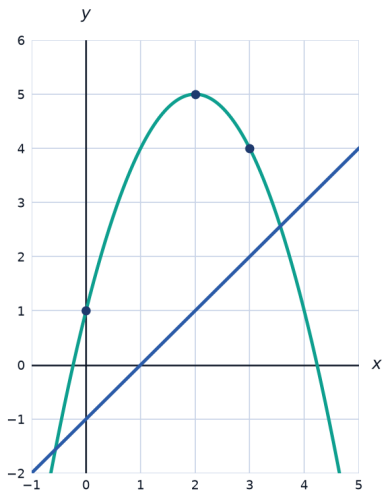
b. 13

c. 6

d. 14

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 4x + 5$ in points and midpoints, evaluate $f(2)$.



a. 13

b. 17

c. 8

d. 17

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25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for line segments and distance consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?

a. 15.5

b. 16.5

c. 17.5

d. 19.5

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For the model $f(x) = 5x + 6$ in review: slopes of lines, evaluate $f(4)$.

a. 26

b. 31

c. 20

d. 30

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

On the coordinate plane, a line with slope $m = 2$ passes through $(0, 7)$. What is its y-coordinate when $x = 1$?

a. 7

b. 11

c. 9

d. 11

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

In ΔABC studying review: angles and angle measure, $m\angle A = 34^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 87°

b. 97°

c. 79°

d. 146°

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A geometric figure scaled by factor $k = 5$ has an original side length of 8 cm. What is the corresponding image length?

a. 35 cm

b. 40 cm

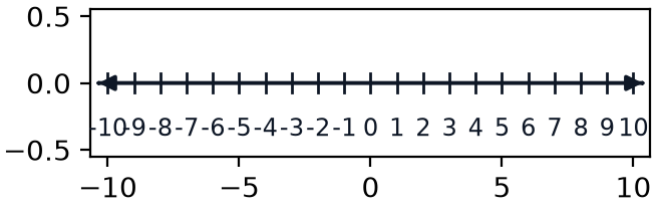
c. 48 cm

d. 200 cm

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30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A student checks that $3, 5, 7, 9$ are the first four terms of a pattern, all odd, and concludes the next term will also be odd. This is an example of:



a. Deductive reasoning

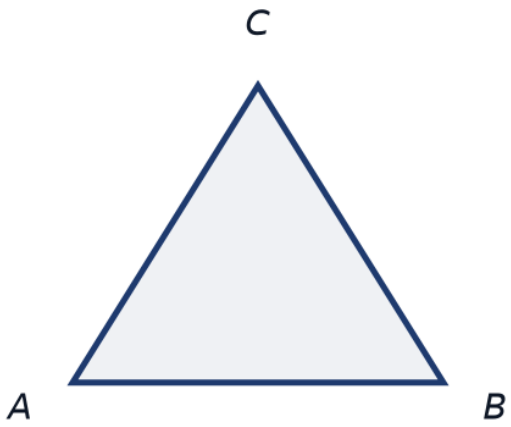
b. Circular reasoning

c. A logical fallacy

d. Inductive reasoning

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The base angles of any isosceles triangle are congruent; this triangle is isosceles, so its base angles are congruent. This is an example of:



a. Circular reasoning

b. Deductive reasoning

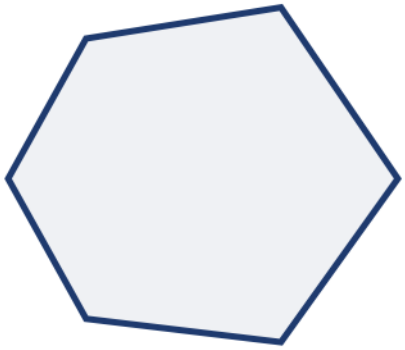
c. Inductive reasoning

d. A logical fallacy

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32 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

A student notices the first three shapes in a sequence are all pentagons and predicts the fourth shape is also a pentagon. This is an example of:



a. A logical fallacy

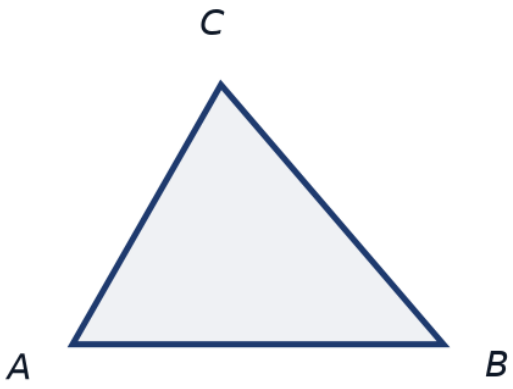
b. Inductive reasoning

c. Circular reasoning

d. Deductive reasoning

33 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

After measuring the angles of five different triangles and finding each sums to 180°, a student concludes all triangles have angle sum 180°. This is an example of:



a. A logical fallacy

b. Circular reasoning

c. Inductive reasoning

d. Deductive reasoning

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34 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

A student measures the diagonals of four different rectangles, finds them equal each time, and concludes all rectangles have equal diagonals. This is an example of:



a. A logical fallacy

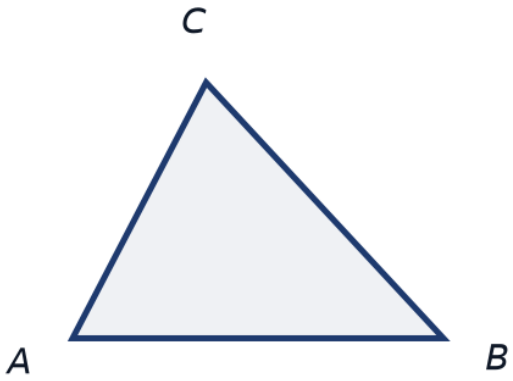
b. Deductive reasoning

c. Inductive reasoning

d. Circular reasoning

35 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

The sum of the angles in any triangle is 180°; this triangle has angles 70° and 60°, so its third angle is 50°. This is an example of:



a. Inductive reasoning

b. Circular reasoning

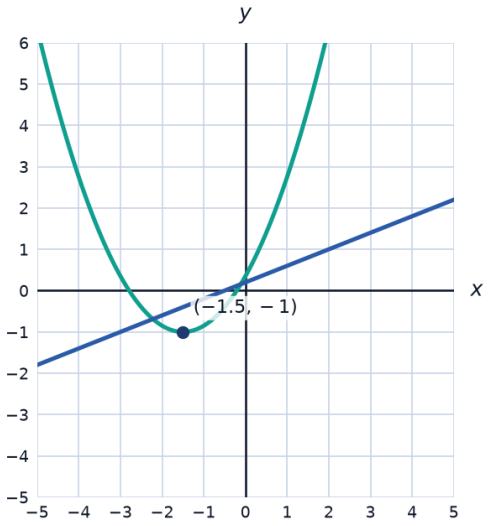
c. A logical fallacy

d. Deductive reasoning

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36 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

Noticing that $1, 4, 9, 16$ are the first four square numbers, a student predicts the next term is 25. This is an example of:



a. Circular reasoning

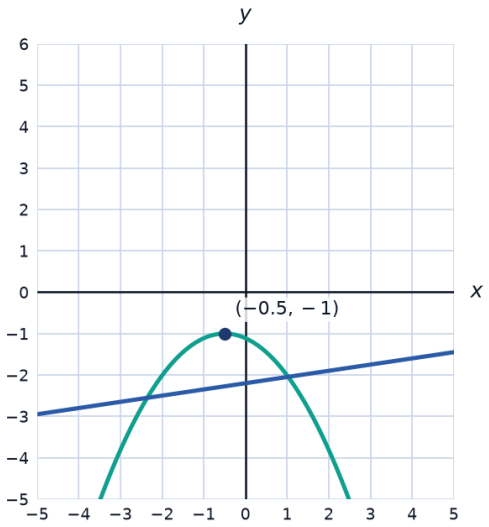
b. A logical fallacy

c. Inductive reasoning

d. Deductive reasoning

37 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

A student tests $n = 1, 2, 3$ in a formula, finds it works each time, and concludes the formula works for all n. This is an example of:



a. Circular reasoning

b. A logical fallacy

c. Inductive reasoning

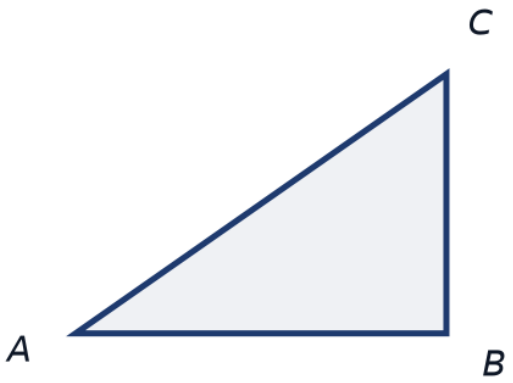
d. Deductive reasoning

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)** • **Cross-Lesson: 7.01 Circles**

What is the contrapositive of the statement: "If A triangle has a right angle, then it is a right triangle."?



a. If It is a right triangle, then a triangle has a right angle.

b. If It is not a right triangle, then a triangle does not have a right angle.

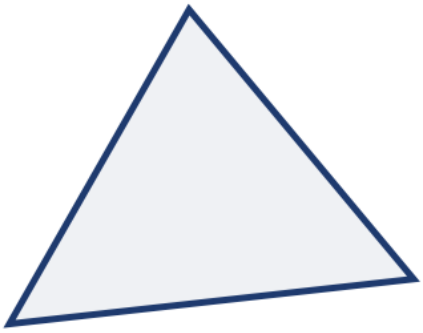
c. If A triangle does not have a right angle, then it is not a right triangle.

d. If A triangle does not have a right angle, then it is a right triangle.

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39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 8.01 Area, Surface Area & Volume

Through any three noncollinear points, there is exactly one ___.



a. line

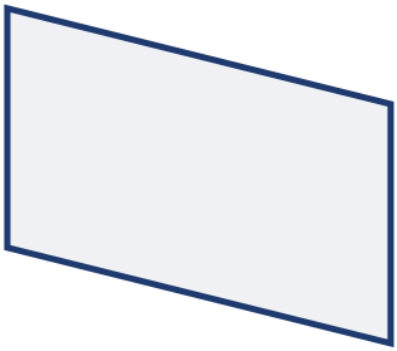
b. segment

c. ray

d. plane

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 9.01 Transformations & Symmetry

If two distinct planes intersect, their intersection is a ___.



a. ray

b. line

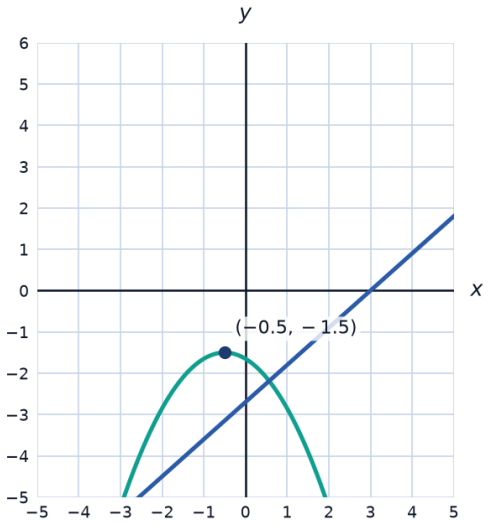
c. point

d. segment

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41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 10.01 Coordinate Geometry

If B is between A and C on a line, then by the Segment Addition Postulate, $AB + BC =$



a. $AC/2$

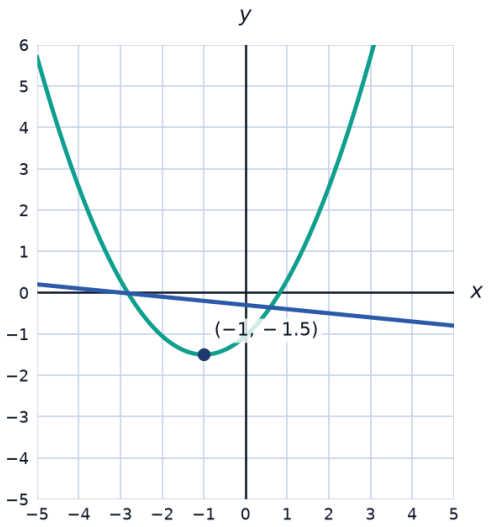
b. $2AC$

c. AC

d. BC

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 11.01 Probability

A line contains at least ___ points.



a. infinitely many

b. one

c. three

d. two

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43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 2.01 Lines & Angles

What is the inverse of the statement: "If A quadrilateral is a parallelogram, then its opposite sides are parallel."?



a. If A quadrilateral is a parallelogram, then its opposite sides are not parallel.

b. If Its opposite sides are parallel, then a quadrilateral is a parallelogram.

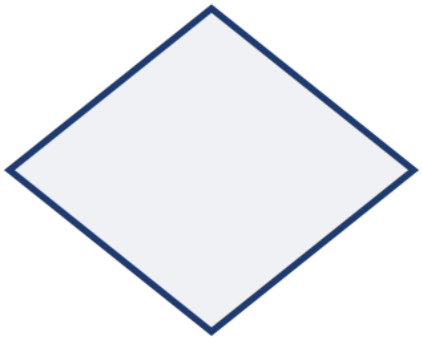
c. If A quadrilateral is not a parallelogram, then its opposite sides are not parallel.

d. If Its opposite sides are not parallel, then a quadrilateral is not a parallelogram.

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44 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)** • **Cross-Lesson: 3.01 Triangles & Congruence**

What is the inverse of the statement: "If A shape is a rhombus, then its diagonals are perpendicular."?



a. If A shape is not a rhombus, then its diagonals are not perpendicular.

b. If Its diagonals are not perpendicular, then a shape is not a rhombus.

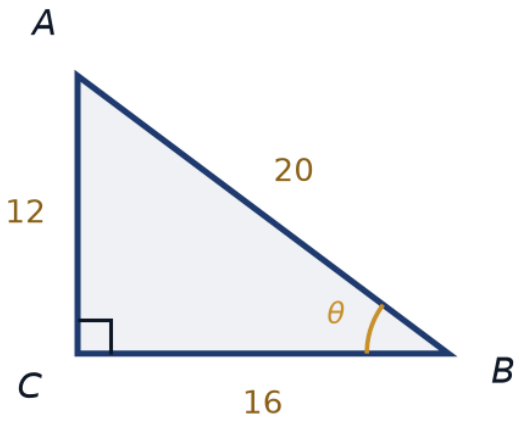
c. If Its diagonals are perpendicular, then a shape is a rhombus.

d. If A shape is a rhombus, then its diagonals are not perpendicular.

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45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.01 Similarity & Proportional Reasoning

What is the converse of the statement: "If A polygon is a triangle, then it has exactly three sides."?



a. If It has exactly three sides, then a polygon is a triangle.

b. If It has exactly three sides, then a polygon is not a triangle.

c. If A polygon is not a triangle, then it does not have exactly three sides.

d. If It does not have exactly three sides, then a polygon is not a triangle.

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